Introduction to Artificial Intelligence

AI as a Science, AI as Engineering

Ashwin Srinivasan

What comes to mind when you hear "Artificial Intelligence"?

Most people today immediately think of

- ChatGPT
- Gemini
- Claude
- Neural Networks
- Deep Learning
- Large Language Models

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Question

Is AI simply another name for Large Language Models?

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$\boxed{\textbf{No.}}$

Artificial Intelligence is a Scientific Discipline

Artificial Intelligence asks questions such as

- What is intelligence?
- How can an agent solve problems?
- How should an agent reason?
- How can an agent learn?
- How should uncertainty be handled?
- How should decisions be made?
- How do intelligent agents cooperate?
- What is creativity?

These questions are scientific questions.

Science versus Engineering

Every mature discipline contains both

1. Scientific theories
2. Engineering tools

Theories explain phenomena.
Tools allow us to investigate, test and exploit those theories.

Example: Physics

Science

- Newton's Laws
- Maxwell's Equations
- Thermodynamics
- Quantum Mechanics

Engineering

- Telescopes
- Particle Accelerators
- Satellites
- Semiconductors

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Observation

Physics is not telescopes.
Physics is the science that explains the universe.

Example: Biology

Science

- Evolution
- Genetics
- Ecology
- Physiology

Engineering

- PCR
- DNA Sequencing
- CRISPR
- Gene Editing

Example: Biology

Science

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Observation

Biology is not CRISPR.
CRISPR is a technology developed using biological knowledge.

Artificial Intelligence

Exactly the same distinction exists.

Science

- Problem Solving
- Knowledge Representation
- Reasoning
- Planning
- Uncertainty
- Learning
- Decision Making

Engineering

- Search algorithms
- Decision Trees
- Neural Networks
- Transformers
- LLMs
- Robotics

The Questions Stay the Same

Engineering changes.
Scientific questions remain.

1950s	Symbolic AI
1970s	Expert Systems
1990s	Probabilistic AI
2000s	Machine Learning
2010s	Deep Learning
2020s	Large Language Models

Each generation developed new engineering tools.
The scientific questions remained remarkably stable.

Capabilities Rather than Algorithms

Instead of asking

"Which program should I implement?"

we should first ask

"What capability are we trying to build?"

and then

"Which algorithms and tools should I use to achieve it?"

The Scientific Landscape of AI

Capability	Scientific Question	Example Engineering Methods
Search	How do we find solutions?	BFS, DFS, A*
Knowledge	How is knowledge represented?	Logic, Knowledge Graphs
Reasoning	How do we infer new facts?	Resolution, Prolog
Uncertainty	How do we reason under uncertainty?	Bayesian Networks, HMMs
Learning	How do systems improve?	Decision Trees, Neural Networks
Planning	How are goals achieved?	STRIPS, PDDL
Perception	How do we interpret sensory input?	CNNs, Vision Transformers
Language	How do we understand language?	Transformers, LLMs
Decision Making	How should actions be selected?	MDPs, Reinforcement Learning
Multi-Agent Systems	How do intelligent agents cooperate?	Game Theory, MARL

Where does Machine Learning fit?

Artificial Intelligence

$\downarrow$

Learning

$\downarrow$

Symbolic Learning

Statistical Learning

Deep Learning

Foundation Models

Where does Machine Learning fit?

Key Scientific Points

Why should machines learn?

What can machines learn?

Key Engineering Point

How should machines learn?

Where does Deep Learning Fit?

Neural networks are one computational approach to learning.

These are neural network implementing one form of machine learning (function approximation), involving artificial interconnected computational units ("neurons") and continuous optimisation that can be used to estimate complex functions many areas:

- Computer Vision
- Speech Recognition
- Natural Language Processing
- Robotics
- Scientific Discovery

But many AI problems require additional capabilities such as

- logical reasoning and consistency
- planning
- explanation
- causal inference

A Broader View of AI

Artificial Intelligence is

the study of what comprises intelligent behaviour.

and

the computational tools we build to investigate and implement that behaviour.

Looking Ahead

Throughout this course we shall continually ask

1. What intelligent capability are we studying?
2. What mathematical principles explain it?
3. What are the latest computational techniques to implement the capabilities we are studying

At the end of the course, if someone asks you: What is the difference between:

- AI and ML?
- ML and DL?
- DL and LLMs?

You should be able to answer coherently